

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q24-2377

Duck Creek 138 kV
January 21, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q24-2377. The Interconnection Customer proposes a 80.0 MW battery energy storage facility interconnecting at the Duck Creek 138 kV substation in the Pennsylvania area. The request was received on January 30, 2024. The study evaluated system intact and single contingency conditions using the regional planning case, including 528 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$13,088,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Lions Rd 345-138 kV Auto	Piney Creek Project-Lions Road 500.0 kV	144.3	381	5.7
Allegheny Hydro No 9-Kirkland Park Substation 138.0 kV	Knott-Primm 138.0 kV	122.8	333	6.4
Phoenix Hydro Project-Neark 138.0 kV	Elizabethton-Tesoro Kenai Cogeneration Plant 138.0 kV	109.6	280	5.0
El Paso-Gilstrap 345.0 kV	Ganso-El Paso 500.0 kV	103.7	1,175	5.6
Skiland-Corning 345.0 kV	Ganso Substation-Handsome Lake 500.0 kV	102.8	961	6.0
Blackrock-Eklutna 345.0 kV	Caldwell-Piney Creek Project 345.0 kV	102.5	1,380	6.4
Bluebonnet-Jones Chapel 230.0 kV	Robison Park-Russell Fork 230.0 kV	102.0	2,635	6.1
Pueblo Xtra-Boca Raton 500.0 kV	Bluebonnet-Jones Chapel 230.0 kV	101.6	3,392	9.0
South Springfield Substation-Knott 138.0 kV	Knott-Double Springs 138.0 kV	101.0	288	7.3
Gilstrap-Ganso 345.0 kV	Ganso-El Paso 500.0 kV	100.7	1,041	6.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Lion Road Substation 345 kV/138 kV Transformer	Replace transformer with higher-rated unit	\$20,448,000	27% (\$5,596,000)
Kirkland Park-Allegheny Hydro No 9 138.0 kV	Rebuild line with higher-capacity conductor	\$6,634,000	28% (\$1,824,000)
Phoenix Hydro Project-Neark 138.0 kV	Reconductor line	\$1,653,000	100% (\$1,653,000)
El Paso-Gilstrap 345.0 kV	Reconductor line	\$20,501,000	1% (\$245,000)
Skiland-Corning 345.0 kV	Reconductor line	\$9,212,000	4% (\$393,000)
Blackrock-Eklutna 345.0 kV	Reconductor line	\$29,536,000	5% (\$1,503,000)
Bluebonnet-Jones Chapel 230.0 kV	Reconductor line	\$14,686,000	2% (\$221,000)
Pueblo Xtra-Boca Raton 500.0 kV	Reconductor line	\$36,868,000	2% (\$547,000)
S Springfield-Knott 138.0 kV	Reconductor line	\$2,852,000	28% (\$799,000)
Gilstrap-Ganso 345.0 kV	Reconductor line	\$11,963,000	3% (\$307,000)
	Total allocated to project		\$13,088,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.